On the Divergence of Calvo and Rotemberg Pricing and the Sustainability of the Standard Paradox of Flexibility Solution at the Zero Lower Bound

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Abstract

The standard New Keynesian literature, notably the Werning (2011) framework, models Zero Lower Bound (ZLB) episodes as finite-duration negative natural interest rate shocks. This paper mathematically demonstrates that the non-linear solution fundamentally diverges between Calvo and Rotemberg pricing environments due to the presence of price dispersion as an independent state variable in the former. In particular, the standard backward-explosive (Werning) solution with approximately zero output gap and zero inflation at the ZLB endpoint cannot mathematically be maintained in the full non-linear Calvo environment. **JEL Classification:** E31, E43, E52

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1 Introduction

In the popular simple New Keynesian approach to liquidity traps, the economy is subjected to a transient preference or discount factor shock that drives the natural rate of interest, r_t^n, into negative territory for a finite duration T. Bound by the zero lower bound on nominal interest rates ($i_t \geq 0$), the central bank cannot lower the policy rate sufficiently to clear the market at the zero-inflation steady state.

1.1 Preceding Works on the Paradox of Flexibility in New Keynesian Models

The foundational framework for analyzing the ZLB in a New Keynesian setting was established by Eggertsson and Woodford (2003), who demonstrated the optimality of forward guidance and history-dependent monetary policy. Building on this, Eggertsson (2010) formalized the "paradox of toil" and the related "paradox of flexibility." The paradox of flexibility posits that as prices become less sticky (i.e., as the price adjustment friction vanishes), the macroeconomic contraction at the ZLB actually worsens, leading to theoretically zero output and infinitely deflationary collapses in the flexible-price limit.

Werning (2011) provided a continuous-time formalization of this phenomenon, sharply illustrating the extreme determinacy of these deflationary trajectories by solving the system backward from a strictly enforced post-ZLB steady state. However, the reliance on a single, globally explosive (while locally stable and bounded) equilibrium path has been heavily criticized. Cochrane (2017) has consistently argued that the extreme predictions of the standard New Keynesian model at the ZLB, including the paradox of flexibility, are not fundamental economic realities but rather artifacts of arbitrary equilibrium selection governed by terminal conditions.

Eggertsson and Singh (2019) further defends against the charge that the Paradox of Flexibility is a log-linearization artifact by adopting an industry-wide labor market instead of an economy-wide labor market. This issue is further discussed in Section 1.4.

1.2 The Log-Linearized Model

To formalize this environment, we assume zero technology growth without loss of generality, which simplifies the analysis. The standard log-linearized New Keynesian model is governed by three equations. In discrete time, the system consists of the dynamic IS equation, the New

Keynesian Phillips Curve (NKPC), and the Monetary Policy (MP) rule (Galí, 2008):

$$x_t = \mathbb{E}_t x_{t+1} - \sigma^{-1}(i_t - \mathbb{E}_t \pi_{t+1} - r_t^n) \tag{1}$$

$$\pi_t = \beta \mathbb{E}_t \pi_{t+1} + \kappa x_t \tag{2}$$

$$i_t = \max(0, r_t^n + \phi_\pi \pi_t + \phi_x x_t) \tag{3}$$

where x_t is the output gap, π_t is inflation, σ is the inverse elasticity of intertemporal substitution, and κ is the slope of the Phillips curve.

Werning's continuous-time counterpart (Werning, 2011) elegantly isolates the ZLB dynamics. Assuming a Taylor rule that perfectly stabilizes the economy outside the ZLB ($t \geq T$), the ZLB period ($0 \leq t < T$ with $i_t = 0$ and $r_t^n = \bar{r} < 0$) is governed by:

$$\dot{x}_t = -\sigma^{-1}\pi_t - \sigma^{-1}\bar{r} \tag{4}$$

$$\dot{\pi}_t = \rho \pi_t - \kappa x_t \tag{5}$$

where ρ is the continuous-time discount rate. As the Calvo parameter $\theta \to 0$ (prices become flexible) (Calvo, 1983), $\kappa \to \infty$. Solving backward from $\pi_T = 0$ and $x_T = 0$, the standard solution predicts that the output gap and inflation collapse to negative infinity during the ZLB.

However, this conventional wisdom masks a severe mathematical inconsistency in the standard (Werning) solution of the Calvo pricing model (Calvo, 1983) that is notably absent in the Rotemberg pricing model (Rotemberg, 1982) when nonlinear constraints regarding price dispersion are considered. There are caveats about price dispersion as a state variable, which is discussed further.

1.3 Preceding Works on Non-linear versus Log-linear Comparisons

There have been papers that consider how the log-linearized solution of a New Keynesian model differ from the full non-linear solution (Gavin et al., 2015; Fernández-Villaverde et al., 2015; Boneva, Braun and Waki, 2016; Miao and Ngo, 2021). This paper differs from these preceding works in that its emphasis is on the paradox of flexibility and is analytical (exact) in its approach, rather than being numerical or simulation-based. Furthermore, since the emphasis is placed on price flexibility, we focus on the Calvo pricing model.

By contrast, Gavin et al. (2015); Boneva, Braun and Waki (2016) focus on the Rotemberg

pricing model. Fernández-Villaverde et al. (2015); Miao and Ngo (2021) do not analyze the flexible-price limit and sets $\theta = 0.75$ (which is the standard empirical value), even though Miao and Ngo (2021) compares the non-linear Calvo model against the non-linear Rotemberg model.

1.4 Industry-wide versus Economy-wide Labor Market: Price Dispersion as an Irrelevant State Variable?

Eggertsson and Singh (2019), building on Woodford (2003), suggests that if an industry-wide labor market instead of an economy-wide labor market, then price dispersion is irrelevant for an aggregate equilibrium, which makes the log-linear approximation very close to the full nonlinear result.

Even without questioning the analysis of Eggertsson and Singh (2019), this paper makes clear that once an economy-wide labor market is restored, the full nonlinear result eliminates the paradox of flexibility in the Calvo model - or at least, the paradox does not arise as in the conventional understanding.

Section 3.2 proves that while the equilibrium outcome is indeed determined without price dispersion involved, as stated in Eggertsson and Singh (2019), it is the very question of equilibrium existence that turns out to be problematic - this is shown in the $\theta \to 0$ limit (flexible-price limit). This arises generically, not constrained to particular parameter values. The intuition behind this result is very simple. If $P_t^*/P_t < 1$ even in the $\theta \to 0$ limit, where P_t^* refers to the price set by symmetric firms able to change their prices, then the CES demand function has output $Y_t^* > Y_t$ in the $\theta \to 0$ limit. But since we are taking the $\theta \to 0$ limit, the CES aggregator mathematics dictates that $Y_t \to Y_t^*$ as $\theta \to 0$. In this sense, $\lim_{\theta \to 0^+} P_t^*/P_t = 1$ in a deflationary episode acts as an additional price dispersion-related constraint. In terms of Eggertsson and Singh (2019), this is an additional constraint on $\lim_{\theta \to 0^+} K_t/F_t = 1$.

1.5 Structure of the Paper

Section 2 discusses why the exact imposition of zero output gap $x_t = 0$ and zero inflation $\pi_t = 0$ at the post-ZLB start point is implausible for non-zero degree of price stickiness θ in the Calvo model.

This still leaves the possibility of $x_t, \pi_t \approx 0$, which is eliminated in the $\theta \to$ limit, discussed in Section 3.1 (economy-wide labor market) and 3.2 (industry-wide or variety-wide labor market, as in Eggertsson and Singh (2019)).

Section B in the appendix discusses the continuous-time version, resulting in the same conclusion in the $\theta \to 0$ limit.

In case of an industry-wide labor market, a resolution is possible, but this requires introducing the wedge between consumption and output in a consumption-only economy, rendering the Calvo model very similar to the Rotemberg model. Otherwise, the Calvo model significantly diverges from the Rotemberg model, and the paradox of flexibility and the standard Werning solution do not hold in the Calvo model.

2 Background: The Price Dispersion Wedge in the Calvo Model for Non-zero Price Stickiness

In the non-linear Calvo framework (Calvo, 1983), a history of non-zero inflation generates price dispersion. Let S_t denote the price dispersion state variable, which aggregates the relative prices of firms stuck at historical price levels (see p. 46 of Galí (2008) for the same definition, with $\alpha = 0$):

$$S_t = \int_0^1 \left(\frac{P_{i,t}}{P_t}\right)^{-\epsilon} di \tag{6}$$

The discrete-time recursive evolution of S_t is given by (see the appendix for more details):

$$S_t = (1-\theta)\left(\frac{P_t^*}{P_t}\right)^{-\epsilon} + \theta \Pi_t^{\epsilon} S_{t-1} = (1-\theta)\left(\frac{1-\theta\Pi_t^{\epsilon-1}}{1-\theta}\right)^{\frac{\epsilon}{\epsilon-1}} + \theta \Pi_t^{\epsilon} S_{t-1} \tag{7}$$

where $\Pi_t = P_t/P_{t-1}$ is gross inflation and P_t^* is the price set by the symmetric firms able to set their prices. During the ZLB period ($0 \leq t < T$), the economy under the standard Werning solution experiences severe deflation ($\Pi_t < 1$), causing S_t to drift strictly above 1. At $t = T$, the economy inherits a strictly positive price dispersion: $S_T > 1$. This can be verified by substituting $\Pi_T = 1$ (of the standard Werning solution):

$$S_T = (1-\theta) + \theta S_{T-1},\ S_{T-1} > 1,\ \theta > 0 \implies S_T > 1 \tag{8}$$

At the aggregate level, the relationship between aggregate output Y_t and aggregate labor N_t is heavily distorted, defined by $Y_t = A_t N_t / S_t$. Consequently, using standard CRRA household preferences with inverse labor supply elasticity φ and scaling parameter χ, the real marginal cost is shifted by this dispersion wedge in case of an economy-wide labor market (see the appendix

for the derivation):

$$MC_t = \chi A_t^{-(1+\varphi)} Y_t^{\sigma+\varphi} S_t^{\varphi} \tag{9}$$

The standard (Werning) solution (Werning, 2011) posits that at $t = T$, the central bank perfectly stabilizes the economy, achieving $Y_t = Y^{ss} \equiv Y$ and $\Pi_t = 1$ indefinitely. However, if $S_T > 1$, S_t will asymptotically decay toward 1 but remains strictly greater than 1 for finite $t > T$. Real marginal cost remains time-varying due to the persistent S_t^{φ} term. This makes it mathematically impossible for firms to optimally choose a constant zero inflation path ($\Pi_t = 1$) while simultaneously clearing the market at steady-state output.

In case of an industry-wide labor market as in Eggertsson and Singh (2019), the real marginal cost argument cannot be used. However, because $S_T > 1$, it is not assured that Y_t calculated from the CES aggregator and the assumed $Y_{i,t}$ equals the assumed Y_t used for calculating each $Y_{i,t}$ from the CES demand function. This is made precise in Section 3.2 in the $\theta \to 0$ limit.

3 The Impossibility of the Standard Flexible-Price Limit in the Non-linear Calvo Model under ZLB: Exact Discrete-Time Approach

However, it may be said that the non-linear impacts of S_t could be so limited such that $\pi_t = 0$ may reasonably be used as an approximation. It may also be argued that even if the non-linear impacts of S_t are significant, they disappear as $\theta \to 0$. The latter is shown not to tbe case - Section 3.1 proves the point in case of an economy-wide labor market, and Section 3.2 in case of an industry-wide (or a variety-wide) labor market (as in Eggertsson and Singh (2019)).

The approach used here is fully nonlinear and exact, and the approximation is only used whenever fully justified, such as in $e^{\zeta x_{T-1}} \approx 1+\zeta x_{T-1}$ where it is already known that $|\zeta x_{T-1}|$ is a small value ($x_{T-1} = -\sigma^{-1}|\bar{r}|$). We do not use the New Keynesian Phillips curve approximation.

The standard Werning solution says $x_t \approx 0$ and $\pi_t \approx 0$ for $t \geq T$, the following pricing equation holds (with details in the appendix, but this is the same equation as in Galí (2008) with $\chi = 1$, $\alpha = 0$):

$$\frac{P^*_{T-1}}{P_{T-1}} \equiv Z_{T-1} = \frac{e^{(1-\sigma)x_{T-1}} e^{\zeta x_{T-1}} e^{\varphi s_{T-1}} + \frac{\beta\theta}{1-\beta\theta}}{e^{(1-\sigma)x_{T-1}} + \frac{\beta\theta}{1-\beta\theta}} = \frac{e^{(1-\sigma+\zeta)x_{T-1}} e^{\varphi s_{T-1}} + \frac{\beta\theta}{1-\beta\theta}}{e^{(1-\sigma)x_{T-1}} + \frac{\beta\theta}{1-\beta\theta}} \tag{10}$$

where $\zeta \equiv \sigma + \varphi$, $s_t \equiv \log S_t$.

In case of Eggertsson and Singh (2019), Z_{T-1} has no dependence on price dispersion since it has an industry-wide labor market (see K_t/F_t in Eggertsson and Singh (2019) for a cross-check)

$$Z_{T-1,ALT} = \frac{e^{(1-\sigma+\zeta)x_{T-1}} + \frac{\beta\theta}{1-\beta\theta}}{e^{(1-\sigma)x_{T-1}} + \frac{\beta\theta}{1-\beta\theta}} \approx 1 - \zeta\sigma^{-1}|\bar{r}| \quad \text{(for small } \theta) \tag{11}$$

where we use the fact that $|x_{T-1}|$ is a small value if $|\bar{r}|$ is a small value.

3.1 Economy-wide Labor Market Case

Substitute $\theta = 0$ and obtain:

$$\lim_{\theta\to 0^+} Z_{T-1} = \frac{e^{(1-\sigma+\zeta)x_{T-1}} e^{\varphi s_{T-1}}}{e^{(1-\sigma)x_{T-1}}} = e^{\zeta x_{T-1}} e^{\varphi s_{T-1}} \approx (1 - \zeta\sigma^{-1}|\bar{r}|) S_{T-1}^{\varphi} \tag{12}$$

Because $S_{T-1} \geq 1$,

$$\lim_{\theta\to 0^+} Z_{T-1} \geq 1 - \zeta\sigma^{-1}|\bar{r}| \tag{13}$$

The aggregate price index formula goes as (see the appendix for derivation - also p. 42 of Galí (2008)):

$$P_t^{1-\epsilon} = (1-\theta)(P_t^*)^{1-\epsilon} + \theta P_{t-1}^{1-\epsilon} \tag{14}$$

Substitute $P_{T-1}^* = kP_{T-1}$ - in case $Z_{T-1} < 1$, $1 - \zeta\sigma^{-1}|\bar{r}| \leq k < 1$:

$$P_{T-1}^{1-\epsilon} = (1-\theta)(kP_{T-1})^{1-\epsilon} + \theta P_{T-2}^{1-\epsilon} \implies 1 = (1-\theta)k^{1-\epsilon} + \theta\left(\frac{P_{T-1}}{P_{T-2}}\right)^{\epsilon-1} \tag{15}$$

$\lim_{\theta\to 0}(1-\theta)k^{1-\epsilon} = k^{1-\epsilon}$, but if $k < 1$ then $k^{1-\epsilon} > 1$ for $\epsilon > 1$, which in turn requires $P_{T-1}/P_{T-2} < 0$ - impossible:

$$\lim_{\theta\to 0} \frac{P_{T-1}^*}{P_{T-1}} < 1 \quad (\epsilon > 1) \implies \lim_{\theta\to 0} \frac{P_{T-1}}{P_{T-2}} < 0 \text{ (contradiction)} \tag{16}$$

Therefore, it must be that $\lim_{\theta\to 0} Z_{T-1} \geq 1$. Indeed, if $\pi_{T-1} < 0$ (deflation at $t = T-1$) then $\lim_{\theta\to 0} Z_{T-1} = 1$.

Assume $\lim_{\theta\to 0} Z_{T-1} = 1$ without assuming $\pi_{T-1} < 0$. Then,

$$\lim_{\theta\to 0} S_{T-1} \approx \left(\frac{1}{1 - \zeta\sigma^{-1}|\bar{r}|}\right)^{1/\varphi} \quad (\lim_{\theta\to 0} Z_{T-1} = 1 \text{ assumed}) \tag{17}$$

However, the fact that price dispersion is greater than 1 as $\theta \to 0$ implies that $\lim_{\theta\to 0} P_{T-2}/P_{T-1} = 0$ since $S_t = \int_0^1 \left(\frac{P_{i,t}}{P_t}\right)^{-\varepsilon} di$:

$$\left(S_t = \int_0^1 \left(\frac{P_{i,t}}{P_t}\right)^{-\epsilon} di\right) \wedge \lim_{\theta\to 0} S_{T-1} > 1 \implies \lim_{\theta\to 0} \frac{P_{T-2}}{P_{T-1}} = 0 \tag{18}$$

Therefore, this implies $\lim_{\theta\to 0} \pi_{T-1} = \infty$ - inflation at $t = T-1$ - the standard Werning solution is wrong - either the terminal condition $x_T, \pi_T = 0$ is not even approximately justified or the solution generates a severe economic oscillation in the nonlinear case as $\theta \to 0$ (flexible-price limit).

3.2 Industry-wide Labor Market Case (as in Eggertsson and Singh (2019))

Recall $Z_{T-1,ALT}$ in Equation (11): $Z_{T-1,ALT} \approx 1 - \zeta\sigma^{-1}|\bar{r}|$ for small θ. Because the demand formula goes as $Y^*_{i,T-1} = (Z_{T-1,ALT})^{-\epsilon} Y_{T-1}$ for flexible-price firms ($1-\theta$ of firms),

$$Y^*_{i,T-1} > Y_{T-1} \tag{19}$$

This create an issue in the $\theta \to 0$ limit:

$$\lim_{\theta\to 0^+} Y_{T-1} = \lim_{\theta\to 0^+} \left((1-\theta)(Y^*_{T-1})^{\frac{\epsilon-1}{\epsilon}} + \theta(Y_{T-1,nf})^{\frac{\epsilon-1}{\epsilon}}\right)^{\frac{\epsilon}{\epsilon-1}} \tag{20}$$

If $\lim_{\theta\to 0^+} Y_{T-1,nf} \neq \infty$, which is the case since there is deflation at $t = T-1$, then:

$$\lim_{\theta\to 0^+} Y_{T-1} = Y^*_{T-1} \tag{21}$$

But this is not possible. Therefore, the standard Werning solution cannot be maintained even with an industry-wide labor market as in Eggertsson and Singh (2019).

3.2.1 A potential resolution?

For the industry-wide labor market case, a convenient shortcut exists to restore standard equilibrium existence - remove the assumption that $C_t = Y_t$ and switch to $C_t \neq Y_t$ as with the conventional Rotemberg model. That is, inflation or deflation causes an additional wedge between consumption and output.

$C_{i,t} = Y_{i,t}$ can still be maintained, but the household utility function should not be a

function of individual $C_{i,t}$ - it should just be a function of C_t. This allows $C_{i,t} > C_t$, while $Y_t = (\int_0^1 C_{i,t}^{(\epsilon-1)/\epsilon} di)^{\epsilon/(\epsilon-1)}$ with $\lim_{\theta \to 0} Y_t = C_{i,t}^*$, with $C_{i,t}^*$ referring to output (and consumption good quantity as well) of a firm that can change its price. This makes the Calvo model more like the Rotemberg model - inflation or deflation somewhat adds aggregate resource costs.

For the question of whether this wedge between consumption and output is conceptually reasonable when an economy features only a consumption good, we should not attempt to answer within the limit of this paper. All that should be done for now is to recognize that both the Calvo model with an industry-wide labor market and Rotemberg model feature the wedge between consumption and output such that $C_t \neq Y_t$.

4 Conclusion

This paper revisits the standard log-linearized analysis of the zero lower bound and the associated paradox of flexibility, demonstrating a critical, frequently overlooked mathematical bifurcation between Calvo and Rotemberg pricing frameworks. The standard solution, characterized by an explosive deflationary collapse and an immediate return to a zero-inflation steady state upon ZLB exit, is structurally incompatible with the non-linear Calvo model. The inherited price dispersion from the deflationary ZLB period creates an endogenous productivity wedge that strictly prevents a smooth, immediate jump to the flexible-price steady state. Furthermore, as the price-adjustment friction vanishes, the massive deflation predicted by the log-linearized solution forces the non-linear Calvo framework into mathematically undefined territory, proving the paradox of flexibility to be an artifact of a wrong terminal condition ($x_T, \pi_T = 0$ rather than an intrinsic economic reality.

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A Derivations of the Non-linear Model Equations

A.1 Household Optimization: Euler Equation and Labor Supply

The representative household seeks to maximize expected lifetime utility:

$$\mathbb{E}_0 \sum_{t=0}^{\infty} \beta^t \left(\frac{C_t^{1-\sigma}}{1-\sigma} - \chi \frac{N_t^{1+\varphi}}{1+\varphi} \right) \tag{22}$$

subject to the sequence of nominal budget constraints:

$$P_t C_t + B_t = (1 + i_{t-1}) B_{t-1} + W_t N_t + \Upsilon_t \tag{23}$$

where C_t is consumption, N_t is labor supply, B_t are one-period nominal risk-free bonds, W_t is the nominal wage, and Υ_t represents aggregate firm profits distributed as dividends.

Let λ_t be the Lagrange multiplier associated with the period t budget constraint. The first-order condition with respect to consumption C_t is:

$$C_t^{-\sigma} - \lambda_t P_t = 0 \implies \lambda_t = \frac{C_t^{-\sigma}}{P_t} \tag{24}$$

The first-order condition with respect to nominal bonds B_t is:

$$-\lambda_t + \beta \mathbb{E}_t \left[\lambda_{t+1} (1 + i_t) \right] = 0 \tag{25}$$

Substituting (24) into (25) yields the standard intertemporal Euler equation:

$$\frac{C_t^{-\sigma}}{P_t} = \beta \mathbb{E}_t \left[\frac{C_{t+1}^{-\sigma}}{P_{t+1}} (1 + i_t) \right] \implies C_t^{-\sigma} = \beta \mathbb{E}_t \left[C_{t+1}^{-\sigma} \frac{1 + i_t}{\Pi_{t+1}} \right] \tag{26}$$

To derive the real wage formula, we take the first-order condition with respect to labor supply N_t:

$$-\chi N_t^{\varphi} + \lambda_t W_t = 0 \tag{27}$$

Substituting the shadow value of wealth $\lambda_t = C_t^{-\sigma} / P_t$ into (27) gives the intra-temporal labor supply condition, equating the marginal rate of substitution to the real wage:

$$\frac{W_t}{P_t} = \frac{\chi N_t^{\varphi}}{C_t^{-\sigma}} \tag{28}$$

To aid the cross-check with Eggertsson and Singh (2019), note that σ in this paper is re-labeled as $1/\bar{\sigma}$ in Eggertsson and Singh (2019), χ is λ in Eggertsson and Singh (2019) (though we set $\chi = 1$ in this paper whenever convenient, since it is a largely irrelevant parameter), $\varphi = 0$ in Eggertsson and Singh (2019), replaced with industry-wide exponent ω, which we set as $\omega = 0$ to allow a cross-check. Note also that we set ξ_t, τ_t^w in Eggertsson and Singh (2019) as $\xi_t = 1$, $\tau_t^w = 0$ - these parameters do not appear in a simplified New Keynesian model.

A.2 The Universal Aggregate Price Index: Final Good Firm and Household Equivalence

The fundamental definition of the aggregate price index, which serves as the identical starting point for both the discrete-time and continuous-time Calvo formulations (as well as the Rotemberg model), is typically derived using a representative, perfectly competitive final good producer. However, it is mathematically isomorphic to derive this directly from the household's optimization problem. Both approaches yield the identical universal price index. This is exactly the same approach and definition as in p. 42 of Galí (2008).

The Final Good Firm Approach: Suppose a final good Y_t is produced by assembling a continuum of differentiated intermediate goods, $Y_{i,t}$ for $i \in [0, 1]$, using a standard Dixit-Stiglitz Constant Elasticity of Substitution (CES) aggregator (Dixit and Stiglitz, 1977):

$$Y_t = \left(\int_0^1 Y_{i,t}^{\frac{\epsilon-1}{\epsilon}} di \right)^{\frac{\epsilon}{\epsilon-1}} \tag{29}$$

where $\epsilon > 1$ is the elasticity of substitution between the intermediate varieties.

The final good firm seeks to maximize its profits by choosing the optimal quantity of each intermediate good $Y_{i,t}$, taking the aggregate price P_t and intermediate prices $P_{i,t}$ as given:

$$\max_{Y_{i,t}} \left(P_t Y_t - \int_0^1 P_{i,t} Y_{i,t} di \right) \tag{30}$$

Taking the first-order condition with respect to $Y_{i,t}$ and applying the chain rule to the CES aggregator yields the downward-sloping demand curve for intermediate variety i:

$$Y_{i,t} = \left(\frac{P_{i,t}}{P_t} \right)^{-\epsilon} Y_t \tag{31}$$

Because the final good sector is perfectly competitive and operates under constant returns to

scale, it earns zero economic profit in equilibrium. Total revenue must equal total expenditure on intermediate inputs:

$$P_t Y_t = \int_0^1 P_{i,t} Y_{i,t} di \tag{32}$$

Substituting the optimal demand curve (31) into this zero-profit condition yields:

$$P_t Y_t = \int_0^1 P_{i,t} \left(\frac{P_{i,t}}{P_t} \right)^{-\epsilon} Y_t di = P_t^{\epsilon} Y_t \int_0^1 P_{i,t}^{1-\epsilon} di \tag{33}$$

Dividing both sides by $P_t^{\epsilon} Y_t$ and raising the result to the power of $\frac{1}{1-\epsilon}$, we arrive at the exact non-linear aggregate price index formula:

$$P_t = \left(\int_0^1 P_{i,t}^{1-\epsilon} di \right)^{\frac{1}{1-\epsilon}} \tag{34}$$

Equivalence with Direct Household Aggregation: It is completely immaterial whether the CES aggregation is performed by a final good firm or directly by the representative household. Suppose instead that there is no final good firm, and the household derives utility directly from a composite consumption good C_t, defined by the same CES aggregator over intermediate consumption varieties $C_{i,t}$:

$$C_t = \left(\int_0^1 C_{i,t}^{\frac{\epsilon-1}{\epsilon}} di \right)^{\frac{\epsilon}{\epsilon-1}} \tag{35}$$

To maximize utility, the household must optimally allocate its total consumption expenditure among the differentiated varieties. This reduces to an expenditure minimization problem: for any desired level of composite consumption C_t, the household seeks to minimize total cost $\int_0^1 P_{i,t} C_{i,t} di$.

Setting up the Lagrangian for this expenditure minimization problem:

$$\mathcal{L} = \int_0^1 P_{i,t} C_{i,t} di + \lambda_t \left[C_t - \left(\int_0^1 C_{i,t}^{\frac{\epsilon-1}{\epsilon}} di \right)^{\frac{\epsilon}{\epsilon-1}} \right] \tag{36}$$

The first-order condition with respect to $C_{i,t}$ yields:

$$P_{i,t} = \lambda_t \left(\int_0^1 C_{j,t}^{\frac{\epsilon-1}{\epsilon}} dj \right)^{\frac{1}{\epsilon-1}} C_{i,t}^{-\frac{1}{\epsilon}} = \lambda_t C_t^{\frac{1}{\epsilon}} C_{i,t}^{-\frac{1}{\epsilon}} \tag{37}$$

Rearranging for $C_{i,t}$, we recover the identical downward-sloping demand curve:

$$C_{i,t} = \left(\frac{P_{i,t}}{\lambda_t}\right)^{-\epsilon} C_t \tag{38}$$

To find the shadow price λ_t, we substitute this demand function back into the definition of total expenditure:

$$\int_0^1 P_{i,t} C_{i,t} di = \int_0^1 P_{i,t} \left(\frac{P_{i,t}}{\lambda_t}\right)^{-\epsilon} C_t di = \lambda_t^{\epsilon} C_t \int_0^1 P_{i,t}^{1-\epsilon} di \tag{39}$$

By definition, the Lagrange multiplier λ_t represents the minimized cost of acquiring one unit of the composite consumption good C_t. Therefore, λ_t is exactly the aggregate price index P_t. Substituting P_t for λ_t and requiring that total expenditure equals $P_t C_t$, we have:

$$P_t C_t = P_t^{\epsilon} C_t \int_0^1 P_{i,t}^{1-\epsilon} di \implies P_t^{1-\epsilon} = \int_0^1 P_{i,t}^{1-\epsilon} di \tag{40}$$

This resolves to the exact same formula as Equation (34). Because the mathematical structure of the CES aggregator is identical whether labeled as a production function Y_t or a consumption aggregator C_t, the resulting aggregate price index P_t and the intermediate demand functions are structurally invariant. The choice between a final good firm and direct household aggregation is purely a modeling convention and has absolutely no bearing on the underlying continuous-time or discrete-time price dynamics.

A.3 Discrete-time Aggregate Price Index Recursion

In the Dixit-Stiglitz framework (Dixit and Stiglitz, 1977), final output Y_t is a CES aggregate of a continuum of intermediate varieties $Y_{i,t}$, indexed by $i \in [0, 1]$, with elasticity of substitution $\epsilon > 1$. The aggregate price index P_t is formally defined as (also derived in the appendix):

$$P_t = \left(\int_0^1 P_{i,t}^{1-\epsilon} di\right)^{\frac{1}{1-\epsilon}} \tag{41}$$

Under Calvo pricing, a randomly selected $1 - \theta$ fraction of intermediate firms are allowed to reset their price optimally to P_t^* in period t. The remaining θ fraction must keep their price from the previous period, $P_{i,t-1}$.

Because the probability of price adjustment is independent of a firm's price history, the dis-

tribution of prices among the non-adjusting firms is exactly the same as the overall distribution of prices in period $t-1$. Therefore, we can decompose the integral in (41) into two components: adjusting firms and non-adjusting firms:

$$P_t^{1-\epsilon} = \int_{\text{adjust}} (P_t^*)^{1-\epsilon} di + \int_{\text{no-adjust}} P_{i,t-1}^{1-\epsilon} di \tag{42}$$

Since all adjusting firms face the same symmetric problem, they all choose the same reset price P_t^*. The measure of this group is $1-\theta$. The measure of the non-adjusting group is θ, and their integral aggregates to the previous period's price index $P_{t-1}^{1-\epsilon}$. This yields the recursive evolution of the aggregate price index:

$$P_t^{1-\epsilon} = (1-\theta)(P_t^*)^{1-\epsilon} + \theta P_{t-1}^{1-\epsilon} \tag{43}$$

Dividing both sides by $P_t^{1-\epsilon}$ gives the inflation dynamics formulation used in the main text:

$$1 = (1-\theta)\left(\frac{P_t^*}{P_t}\right)^{1-\epsilon} + \theta \Pi_t^{\epsilon-1} \tag{44}$$

A.4 Aggregate Production Function and the Price Dispersion Wedge

Firm i produces using the linear technology $Y_{i,t} = A_t N_{i,t}$. The labor demanded by firm i is thus $N_{i,t} = Y_{i,t}/A_t$. Aggregate labor is the integral of individual labor demands:

$$N_t = \int_0^1 N_{i,t} di = \int_0^1 \frac{Y_{i,t}}{A_t} di \tag{45}$$

The downward-sloping demand curve for variety i is given by $Y_{i,t} = \left(\frac{P_{i,t}}{P_t}\right)^{-\epsilon} Y_t$. Substituting this into the aggregate labor equation gives:

$$N_t = \int_0^1 \frac{1}{A_t}\left(\frac{P_{i,t}}{P_t}\right)^{-\epsilon} Y_t di = \frac{Y_t}{A_t}\int_0^1 \left(\frac{P_{i,t}}{P_t}\right)^{-\epsilon} di \tag{46}$$

Defining the price dispersion wedge as $S_t = \int_0^1 \left(\frac{P_{i,t}}{P_t}\right)^{-\epsilon} di$, we obtain the aggregate production function linking aggregate output to aggregate labor:

$$N_t = \frac{Y_t S_t}{A_t} \implies Y_t = \frac{A_t N_t}{S_t} \tag{47}$$

A.5 Discrete-time Price Dispersion Recursion (S_t and $Z_t = P_t^*/P_t$)

Using the same logic as the price index decomposition, we split the integral for the price dispersion variable S_t into adjusting and non-adjusting firms:

$$S_t = (1-\theta)\left(\frac{P_t^*}{P_t}\right)^{-\epsilon} + \theta \int_0^1 \left(\frac{P_{i,t-1}}{P_t}\right)^{-\epsilon} di \tag{48}$$

Multiply and divide the second term by $P_{t-1}^{-\epsilon}$:

$$S_t = (1-\theta)\left(\frac{P_t^*}{P_t}\right)^{-\epsilon} + \theta \left(\frac{P_{t-1}}{P_t}\right)^{-\epsilon} \int_0^1 \left(\frac{P_{i,t-1}}{P_{t-1}}\right)^{-\epsilon} di \tag{49}$$

Recognizing that the integral is simply S_{t-1} and that $P_t/P_{t-1} = \Pi_t$, we get:

$$S_t = (1-\theta)\left(\frac{P_t^*}{P_t}\right)^{-\epsilon} + \theta \Pi_t^{\epsilon} S_{t-1} \tag{50}$$

To eliminate the relative reset price P_t^*/P_t, we rearrange the aggregate price index equation (for cross-check, see K_t/F_t in Eggertsson and Singh (2019), with α re-named to ϵ and ϵ re-named to α and $\omega = 1$):

$$1 = (1-\theta)\left(\frac{P_t^*}{P_t}\right)^{1-\epsilon} + \theta \Pi_t^{\epsilon-1} \implies Z_t \equiv \frac{P_t^*}{P_t} = \left(\frac{1-\theta\Pi_t^{\epsilon-1}}{1-\theta}\right)^{\frac{1}{1-\epsilon}} \tag{51}$$

Substituting this expression for the relative reset price back into (50) yields the final recursive law of motion for price dispersion:

$$S_t = (1-\theta)\left(\frac{1-\theta\Pi_t^{\epsilon-1}}{1-\theta}\right)^{\frac{\epsilon}{\epsilon-1}} + \theta \Pi_t^{\epsilon} S_{t-1} \tag{52}$$

A.6 Continuous-Time Aggregate Price Index and Inflation

In the Dixit-Stiglitz framework, the aggregate price index P_t is defined by the CES aggregator over a continuum of intermediate varieties $i \in [0,1]$:

$$P_t = \left(\int_0^1 P_{i,t}^{1-\epsilon} di\right)^{\frac{1}{1-\epsilon}} \implies P_t^{1-\epsilon} = \int_0^1 P_{i,t}^{1-\epsilon} di \tag{53}$$

Let $1-\theta$ denote the instantaneous Poisson arrival rate of price adjustment opportunities. Over a vanishingly small time interval Δt, a fraction $(1-\theta)\Delta t$ of firms are drawn to reset their

price. Because adjusting firms face a symmetric problem, they all choose the identical optimal reset price P_t^*. The remaining $1-(1-\theta)\Delta t$ fraction of firms must keep their existing prices.

The evolution of the price index over this interval Δt is:

$$P_{t+\Delta t}^{1-\epsilon} = ((1-\theta)\Delta t)(P_t^*)^{1-\epsilon} + (1-(1-\theta)\Delta t)P_t^{1-\epsilon} \tag{54}$$

Subtracting $P_t^{1-\epsilon}$ from both sides and dividing by the time increment Δt yields:

$$\frac{P_{t+\Delta t}^{1-\epsilon} - P_t^{1-\epsilon}}{\Delta t} = (1-\theta)\left[(P_t^*)^{1-\epsilon} - P_t^{1-\epsilon}\right] \tag{55}$$

Taking the continuous-time limit as $\Delta t \to 0$, the left-hand side converges to the time derivative of $P_t^{1-\epsilon}$:

$$\frac{d}{dt}\left(P_t^{1-\epsilon}\right) = (1-\theta)\left[(P_t^*)^{1-\epsilon} - P_t^{1-\epsilon}\right] \tag{56}$$

Applying the chain rule to the left-hand side, we have $\frac{d}{dt}\left(P_t^{1-\epsilon}\right) = (1-\epsilon)P_t^{-\epsilon}\dot{P}_t$. Substituting this back into (56) and dividing the entire equation by $P_t^{1-\epsilon}$ gives:

$$(1-\epsilon)\frac{\dot{P}_t}{P_t} = (1-\theta)\left[\left(\frac{P_t^*}{P_t}\right)^{1-\epsilon} - 1\right] \tag{57}$$

Recognizing that instantaneous inflation is defined as $\pi_t = \frac{\dot{P}_t}{P_t}$, we arrive at the exact non-linear continuous-time relationship between aggregate inflation and the relative reset price:

$$\pi_t = \frac{1-\theta}{1-\epsilon}\left[\left(\frac{P_t^*}{P_t}\right)^{1-\epsilon} - 1\right] \tag{58}$$

A.7 Continuous-Time Price Dispersion Recursion

The price dispersion state variable is defined as $S_t = \int_0^1 \left(\frac{P_{i,t}}{P_t}\right)^{-\epsilon} di$. It is mathematically convenient to factor out the aggregate price index P_t, which is independent of the firm index i, to define an unscaled absolute price dispersion measure, Δ_t:

$$\Delta_t = \int_0^1 P_{i,t}^{-\epsilon} di \implies S_t = P_t^{\epsilon}\Delta_t \tag{59}$$

We first determine the continuous-time evolution of Δ_t. Using the same Poisson arrival logic as above, over a small interval Δt, a fraction $(1-\theta)\Delta t$ of firms reset their price to P_t^*, while

$1-(1-\theta)\Delta t$ keep their historical prices. The evolution of Δ_t is:

$$\Delta_{t+\Delta t} = ((1-\theta)\Delta t)(P_t^*)^{-\epsilon} + (1-(1-\theta)\Delta t)\Delta_t \tag{60}$$

Subtracting Δ_t, dividing by Δt, and taking the limit as $\Delta t \to 0$ yields the continuous-time derivative:

$$\dot{\Delta}_t = (1-\theta)\left[(P_t^*)^{-\epsilon} - \Delta_t\right] \tag{61}$$

Now, we differentiate the scaled price dispersion state variable $S_t = P_t^\epsilon \Delta_t$ with respect to time using the product rule:

$$\dot{S}_t = \epsilon P_t^{\epsilon-1}\dot{P}_t\Delta_t + P_t^\epsilon \dot{\Delta}_t \tag{62}$$

We can rewrite the first term by multiplying and dividing by P_t, noting that $\frac{\dot{P}_t}{P_t} = \pi_t$ and $P_t^\epsilon \Delta_t = S_t$:

$$\epsilon P_t^{\epsilon-1}\dot{P}_t\Delta_t = \epsilon\left(\frac{\dot{P}_t}{P_t}\right)P_t^\epsilon\Delta_t = \epsilon\pi_t S_t \tag{63}$$

Next, we substitute the expression for $\dot{\Delta}_t$ from (61) into the second term:

$$P_t^\epsilon\dot{\Delta}_t = P_t^\epsilon\left((1-\theta)(P_t^*)^{-\epsilon} - (1-\theta)\Delta_t\right) = (1-\theta)\left(\frac{P_t^*}{P_t}\right)^{-\epsilon} - (1-\theta)S_t \tag{64}$$

Combining these differentiated components yields the exact non-linear continuous-time law of motion for price dispersion:

$$\dot{S}_t = (1-\theta)\left(\frac{P_t^*}{P_t}\right)^{-\epsilon} - (1-\theta)S_t + \epsilon\pi_t S_t \tag{65}$$

This confirms the structural equation utilized in the main text to prove the explosive trajectory of price dispersion under the paradox of flexibility.

A.8 Real Marginal Cost (MC_t)

In a Calvo model equilibrium, aggregate consumption equals aggregate output ($C_t = Y_t$). Substituting this market clearing condition and the aggregate production function $N_t = \frac{Y_t S_t}{A_t}$ into the intra-temporal labor supply equation (28) yields:

$$\frac{W_t}{P_t} = \frac{\chi\left(\frac{Y_t S_t}{A_t}\right)^\varphi}{Y_t^{-\sigma}} = \chi A_t^{-\varphi} Y_t^{\sigma+\varphi} S_t^\varphi \tag{66}$$

Real marginal cost is defined as the real wage divided by the marginal product of labor (A_t):

$$MC_t = \frac{W_t/P_t}{A_t} = \frac{\chi A_t^{-\varphi} Y_t^{\sigma+\varphi} S_t^{\varphi}}{A_t} = \chi A_t^{-(1+\varphi)} Y_t^{\sigma+\varphi} S_t^{\varphi} \tag{67}$$

Let $\chi, A_t = 1$ for simplification without loss of generality. Let elasticity $\zeta = \sigma + \varphi$ and output gap $x_t = \ln(Y_t/Y)$ where $Y \equiv Y_{ss}$ is steady-state output. Then,

$$MC_t = \frac{\epsilon - 1}{\epsilon} e^{\zeta x_t} e^{\varphi s_t} \tag{68}$$

where we used the fact that steady-state $MC \equiv MC_{ss} = Y^{\sigma+\varphi} = (\epsilon - 1)/\epsilon$ and $s_t \equiv \ln S_t$.

A.9 Calvo-Pricing Firm Optimization Problem

Firms face Calvo pricing, where a randomly selected fraction $(1 - \theta)$ of firms can reset their price in a given period. A resetting firm chooses the optimal nominal price P_t^* to maximize the expected discounted sum of future real profits (see p. 44-45 of Galí (2008), and note that since $\alpha = 0$, real marginal cost MC_t can be used for the total cost evaluation):

$$\max_{P_t^*} \mathbb{E}_t \sum_{k=0}^{\infty} \theta^k Q_{t,t+k} \left[P_t^* Y_{t+k|t} - P_{t+k} MC_{t+k} Y_{t+k|t} \right] \tag{69}$$

subject to the demand curve $Y_{t+k|t} = \left(\frac{P_t^*}{P_{t+k}}\right)^{-\varepsilon} Y_{t+k}$, where $Q_{t,t+k}$ is the stochastic discount factor.

Substituting the demand curve into the objective function yields:

$$\max_{P_t^*} \mathbb{E}_t \sum_{k=0}^{\infty} \theta^k Q_{t,t+k} \left[(P_t^*)^{1-\varepsilon} P_{t+k}^{\varepsilon} Y_{t+k} - (P_t^*)^{-\varepsilon} P_{t+k}^{1+\varepsilon} MC_{t+k} Y_{t+k} \right] \tag{70}$$

Taking the first-order condition with respect to P_t^* and setting it to zero gives:

$$\mathbb{E}_t \sum_{k=0}^{\infty} \theta^k Q_{t,t+k} \left[(1-\varepsilon)(P_t^*)^{-\varepsilon} P_{t+k}^{\varepsilon} Y_{t+k} + \varepsilon (P_t^*)^{-\varepsilon-1} P_{t+k}^{1+\varepsilon} MC_{t+k} Y_{t+k} \right] = 0 \tag{71}$$

Rearranging terms to isolate P_t^* yields:

$$P_t^* \mathbb{E}_t \sum_{k=0}^{\infty} \theta^k Q_{t,t+k} P_{t+k}^{\varepsilon} Y_{t+k} = \frac{\varepsilon}{\varepsilon - 1} \mathbb{E}_t \sum_{k=0}^{\infty} \theta^k Q_{t,t+k} P_{t+k}^{1+\varepsilon} MC_{t+k} Y_{t+k} \tag{72}$$

To express this in terms of the optimal relative reset price $Z_t \equiv P_t^*/P_t$, we divide both

sides by P_t. Assuming the nominal stochastic discount factor takes the standard form $Q_{t,t+k} = \beta^k (Y_{t+k}/Y_t)^{-\sigma}(P_t/P_{t+k})$ (see p. 44 of Galí (2008)), we can rewrite the expression in terms of cumulative future inflation $\Pi_{t,t+k} \equiv P_{t+k}/P_t$:

$$Z_t = \frac{\varepsilon}{\varepsilon - 1} \frac{\mathbb{E}_t \sum_{k=0}^{\infty} (\beta\theta)^k \Pi_{t,t+k}^{\varepsilon} Y_{t+k}^{1-\sigma} MC_{t+k}}{\mathbb{E}_t \sum_{k=0}^{\infty} (\beta\theta)^k \Pi_{t,t+k}^{\varepsilon-1} Y_{t+k}^{1-\sigma}} \tag{73}$$

This naturally condenses into the recursive exact optimal relative reset price:

$$Z_t = \frac{\varepsilon}{\varepsilon - 1} \frac{V_{1,t}}{V_{2,t}} \tag{74}$$

where the recursive auxiliary variables are defined as:

$$V_{1,t} = Y_t^{1-\sigma} MC_t + \beta\theta \mathbb{E}_t \left[\Pi_{t+1}^{\varepsilon} V_{1,t+1} \right] \tag{75}$$

$$V_{2,t} = Y_t^{1-\sigma} + \beta\theta \mathbb{E}_t \left[\Pi_{t+1}^{\varepsilon-1} V_{2,t+1} \right] \tag{76}$$

with $\Pi_{t+1} \equiv P_{t+1}/P_t$ being the gross inflation rate.

For a cross-check with Eggertsson and Singh (2019), note that $\epsilon V_{1,t}/(\epsilon - 1)$ and $V_{2,t}$ are K_t and F_t in Eggertsson and Singh (2019). Furthermore, since Eggertsson and Singh (2019) adopts an industry-wide labor market, MC_t has no dependence on price dispersion S_t and therefore, $MC_t = Y_t^{\sigma}$ in Eggertsson and Singh (2019) with $\varphi = 0$, $\chi, A_t = 1$, $Y_t = C_t$ (variables all following this paper's notation).

Assume that from $t \geq T$, the economy perfectly stabilizes. Therefore, for all $t \geq T$, output is at its steady state ($x_t = 0 \implies Y_t = Y \equiv Y_{ss}$) and inflation is zero ($\Pi_t = 1$). Under these conditions, the future infinite sums converge to exact constants:

$$V_{1,T} = \frac{Y^{1-\sigma} \left(\frac{\varepsilon - 1}{\varepsilon} \right)}{1 - \beta\theta} \tag{77}$$

$$V_{2,T} = \frac{Y^{1-\sigma}}{1 - \beta\theta} \tag{78}$$

At $t = T - 1$, the exact recursive variables evaluate to:

$$V_{1,T-1} = Y_{T-1}^{1-\sigma} MC_{T-1} + \beta\theta \Pi_T^{\varepsilon} V_{1,T} = Y_{T-1}^{1-\sigma} MC_{T-1} + \frac{\beta\theta}{1 - \beta\theta} \frac{\epsilon - 1}{\epsilon} Y^{1-\sigma} \tag{79}$$

$$V_{2,T-1} = Y_{T-1}^{1-\sigma} + \beta\theta \Pi_T^{\varepsilon-1} V_{2,T} = Y_{T-1}^{1-\sigma} + \frac{\beta\theta}{1 - \beta\theta} Y^{1-\sigma} \tag{80}$$

Recognizing that $Y_{T-1} = Ye^{x_{T-1}}$ and $MC_{T-1} = \frac{\varepsilon-1}{\varepsilon}e^{\zeta x_{T-1}}e^{\varphi s_{T-1}}$, we plug them into the optimal relative reset price Z_{T-1}.

$$\frac{P^*_{T-1}}{P_{T-1}} \equiv Z_{T-1} = \frac{e^{(1-\sigma)x_{T-1}}e^{\zeta x_{T-1}}e^{\varphi s_{T-1}} + \frac{\beta\theta}{1-\beta\theta}}{e^{(1-\sigma)x_{T-1}} + \frac{\beta\theta}{1-\beta\theta}} = \frac{e^{(1-\sigma+\zeta)x_{T-1}}e^{\varphi s_{T-1}} + \frac{\beta\theta}{1-\beta\theta}}{e^{(1-\sigma)x_{T-1}} + \frac{\beta\theta}{1-\beta\theta}} \tag{81}$$

where $s_t = \ln S_t$.

B The Impossibility of the Standard Flexible-Price Limit in the Non-linear Calvo Model under ZLB: Continuous-Time Approach

In continuous time, letting $1-\theta$ represent the instantaneous arrival rate of price adjustment, the exact non-linear relationship between aggregate inflation π_t and the optimal relative reset price (derived in the appendix with details) is:

$$\pi_t = \frac{1-\theta}{1-\epsilon}\left[\left(\frac{P^*_t}{P_t}\right)^{1-\epsilon} - 1\right] \tag{82}$$

By rearranging this, we can isolate the optimal relative reset price:

$$\left(\frac{P^*_t}{P_t}\right)^{-\epsilon} = \left(1 - \frac{\epsilon-1}{1-\theta}\pi_t\right)^{\frac{\epsilon}{\epsilon-1}} \tag{83}$$

The state variable $S_t = \int_0^1 (P_{i,t}/P_t)^{-\epsilon} di$ evolves according to the differential equation (derived in the appendix with details):

$$\dot{S}_t = (1-\theta)\left(\frac{P^*_t}{P_t}\right)^{-\epsilon} - (1-\theta)S_t + \epsilon\pi_t S_t \tag{84}$$

Substituting Equation (83) into (84), we obtain the exact non-linear law of motion for price dispersion as a function of inflation:

$$\dot{S}_t = (1-\theta)\left(1 + \frac{\epsilon-1}{1-\theta}|\pi_t|\right)^{\frac{\epsilon}{\epsilon-1}} - (1-\theta)S_t - \epsilon|\pi_t|S_t \tag{85}$$

where we substitute $\pi_t = -|\pi_t|$ to explicitly evaluate the deflationary environment.

$$\lim_{\theta \to 0^+} \dot{S}_t = (1 + (\epsilon - 1)|\pi_t|)^{\frac{\epsilon}{\epsilon-1}} - S_t - \epsilon|\pi_t|S_t \tag{86}$$

We now evaluate this system under the continuous-time log-linearized solution proposed by the paradox of flexibility. In the continuous-time limit (and especially around $\pi_t = 0$ at $t = T$), the ZLB period ($i_t = 0$, $r_t^n = \bar{r} < 0$) is governed by the differential equations:

$$\dot{x}_t = -\sigma^{-1}\pi_t - \sigma^{-1}\bar{r} \tag{87}$$

$$\dot{\pi}_t = \rho\pi_t - \kappa x_t \tag{88}$$

where ρ is the continuous-time discount rate and the NKPC slope is $\kappa \approx \frac{1}{\theta}$ for small θ.

The eigenvalues of this continuous dynamical system are $\lambda_{1,2} = \frac{\rho \pm \sqrt{\rho^2 + 4\sigma^{-1}\kappa}}{2}$. As prices become flexible ($\theta \to 0$ and $\kappa \to \infty$), the roots diverge symmetrically: $\lambda \approx \pm\sqrt{\frac{\kappa}{\sigma}} \approx \pm\sqrt{\frac{1}{\sigma\theta}}$.

Solving backward from the terminal condition $\pi_T = 0$ and $x_T = 0$, the trajectory for $t < T$ is dominated by the unstable root. The continuous-time log-linearized solution takes the form:

$$\pi_t \propto -\exp\left(\sqrt{\frac{1}{\sigma\theta}}(T - t)\right) \tag{89}$$

Translating this to gross inflation, $\Pi_t = \exp(\pi_t)$, we find a double-exponential collapse:

$$\Pi_t \approx \exp\left[-C\exp\left(\frac{\tilde{c}}{\sqrt{\theta}}(T - t)\right)\right] \tag{90}$$

where C and $\tilde{c}$ are positive constants.

As $\theta \to 0$, the deflationary rate $|\pi_t|$ diverges exponentially to infinity. We must determine the dominant term in Equation (85), assuming that S_t stabilizes to small value (proof by contradiction) as ZLB approaches its endpoint $t \to T$. The first term, representing the dispersion introduced by newly adjusting firms, grows at the rate of $|\pi_t|^{\frac{\epsilon}{\epsilon-1}}$. Because the elasticity of substitution $\epsilon > 1$, the exponent $\frac{\epsilon}{\epsilon-1} > 1$. Therefore, this term scales super-linearly with deflation.

The negative term pulling dispersion down is $-\epsilon|\pi_t|S_t$, which scales only linearly with deflation. Consequently, as $|\pi_t| \to \infty$, the super-linear positive term massively dominates the linear

negative term:

$$\lim_{\theta \to 0^+, t \to T^-} \dot{S}_t = +\infty \tag{91}$$

The standard solution forces the derivative of price dispersion to positive infinity. By the time the ZLB period concludes at $t = T$, the economy inherits an infinitely large price dispersion state variable ($S_T \to \infty$). Because aggregate supply is defined by $Y_t = A_t N_t / S_t$ in case of an economy-wide labor market, an infinite price dispersion drives real output to zero for any given labor supply. In case of an industry-wide (or variety-wide) labor market, price dispersion S_t itself suggests $Y_t^* > Y_t$ (See Section 3.2 for detailed arguments), so either the wedge between consumption and output needs to be introduced as in the Rotembberg model or the standard Werning solution does not work.

Note that this result assumes the log-linearized equations that rely on S_t stabilizing to 1. Since this is not satisfied - a contradiction - we can conclude that the standard Werning solution cannot hold.